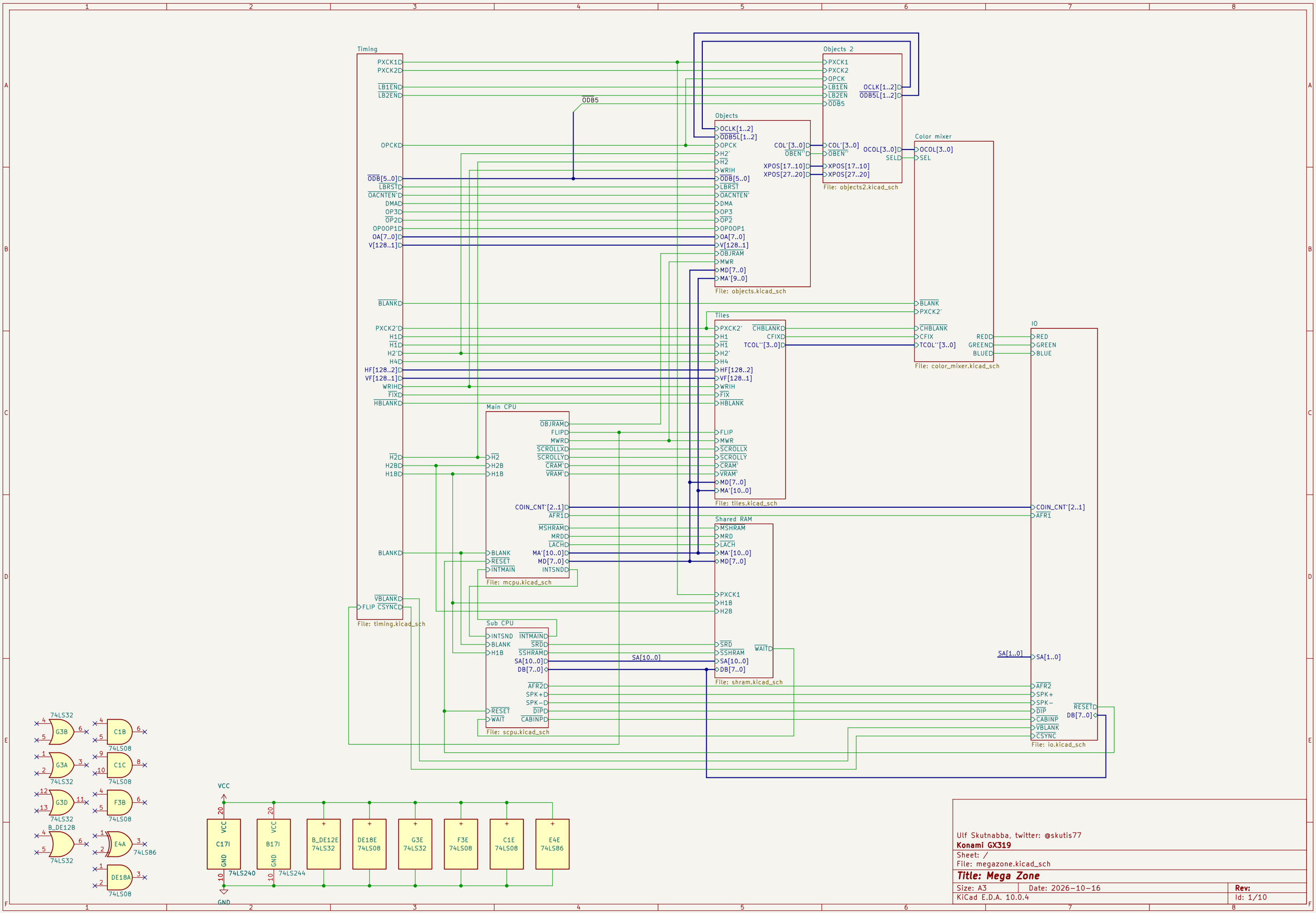


Ulf Skutnabba, twitter: @skutis77

Konami GX319

Sheet: /

File: megazone.kicad_sch

Title: Mega Zone

Size: A3	Date: 2026-10-16	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/10

CFIX and CHBLANK are delayed 4 pixels relative to their source signals FIX and HBLANK.

Konami GX319	
Sheet: /Tiles/	
File: tiles.kicad_sch	
Title: Mega Zone	
Size: A3	Date: 2026-10-16
KiCad E.D.A. 10.0.4	Id: 2/10
	Rev:

OBJ RAM
Buffer 2

E14A
SRAM1K4B

E15A
SRAM1K4B

200ns

E12A
74LS244

D11
74LS273

D13
74LS283

D12
74LS283

D7B
74LS20

E5
74LS273

flip flops

Byte 1: Pos Y

Byte 0, attributes

E6A
74LS174

C3
74LS273

Byte 2: Sprite code

D6
74LS377

74LS32

74LS86

D1A
27C64

D3A
27C64

E3A
27C64

E1A
27C64

C5
083

C6A
PROM256W4B

DE19A
74LS10

74LS04

74LS00

74LS32

DE19B
74LS10

E16
74LS157

F16
74LS157

F14A
SRAM1K4B

F15A
SRAM1K4B

OBJ RAM
Buffer 1

E13
74LS374

3-state
flip flops

G16
74LS245

buffers

74LS08

Byte 3: X Pos

When the corresponding line counter is enabled ,ODB5L1 or ODB5L2, the x positions go from 0..15, drawing every pixel of the 16 x 16 sprite object to the active line buffer for drawing. When the line buffer toggles after LBRST goes high the pixels are read and the x positions go from 0..239 during scroll/obj layers active.

A3
74LS161

A2
74LS161

B2
74LS161

B3
74LS161

counter

C6B
PROM256W4B

E6B
74LS174

DE19D
74LS10

D4E
74LS86

G5E
74LS32

D7C
74LS20

D1B

D3B

E3B

E1B

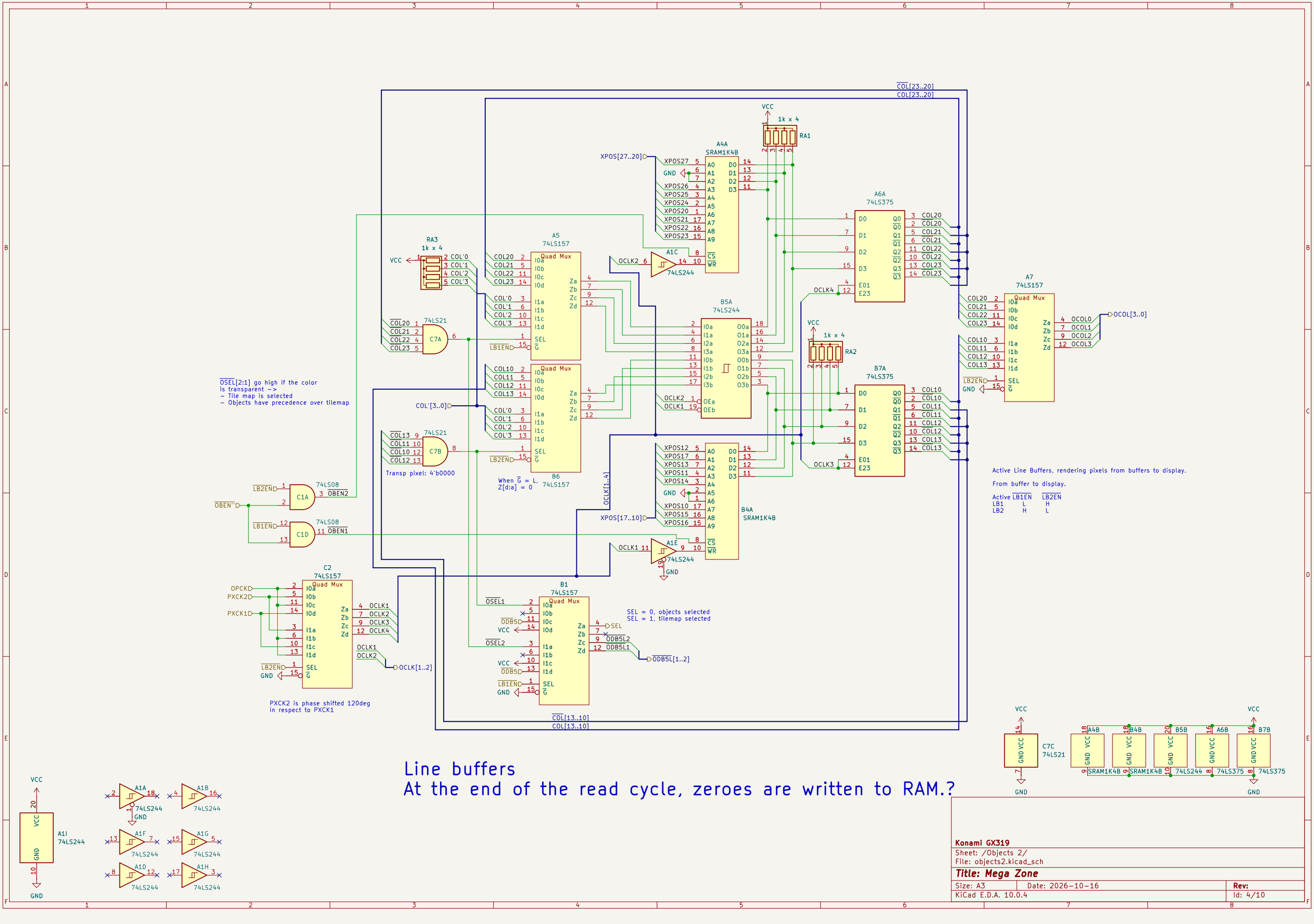
27C64

- DM goes from 0..240 during DMA, data written. OBJ RAM is copied from buffer 1 to buffer 2 during DMA. DM over 144 is redundant, A8 and A9 are connected to GND.
- DM goes from 0..143 otherwise

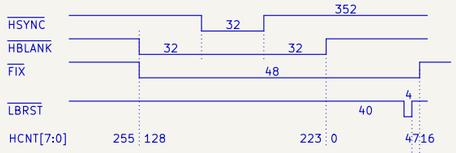
A total of 144/4 = 36 sprites on screen at a time.

A0 V1
A1 V2
A2 V4
A3 px0
A4 px1
A5 V8
A[12:6] Code

Konami GX319	
Sheet: /Objects/	
File: objects.kicad_sch	
Title: Mega Zone	
Size: A3	Date: 2026-10-16
KiCad E.D.A. 10.0.4	Id: 3/10
	Rev:


COL[23..20]
COL[23..20]
VCC
1k x 4
RA1
A4A
SRAM1K4B
XPOS[27..20]
XPOS27
GND
XPOS26
XPOS25
XPOS24
XPOS20
XPOS21
XPOS22
XPOS23
A1C
OCLK2
74LS244
A6A
74LS375
OCLK4
A5
74LS157
Quad Mux
RA3
1k x 4
COL'0
COL'1
COL'2
COL'3
74LS21
C7A
LB1END
B5A
74LS244
VCC
1k x 4
RA2
B7A
74LS375
OCLK3
A7
74LS157
Quad Mux
OCOL[3..0]
LB2END
GND
OSEL[2:1] go high if the color
is transparent —>
– Tile map is selected
– Objects have precedence over tilemap
COL'[3..0]
74LS21
C7B
Transp pixel: 4'b0000
B6
74LS157
When G = L,
Z[d:a] = 0
XPOS12
XPOS17
XPOS13
XPOS11
XPOS14
XPOS10
XPOS15
XPOS16
XPOS[17..10]
B4A
SRAM1K4B
OCLK[1..4]
A1E
74LS244
OCLK1
74LS08
C1A
C1D
OBEN2
OBEN1
OBEN'
C2
74LS157
OPCK
PXCK2
PXCK1
OCLK1
OCLK2
OCLK3
OCLK4
OCLK[1..2]
PXCK2 is phase shifted 120deg
in respect to PXCK1
B1
74LS157
OSEL1
OSEL2
ODB5
ODB5L2
ODB5L1
ODB5L[1..2]
SEL = 0, objects selected
SEL = 1, tilemap selected
COL[13..10]
COL[13..10]
Active Line Buffers, rendering pixels from buffers to display.
From buffer to display.
Active LB1EN LB2EN
LB1 L H
LB2 H L
Line buffers
At the end of the read cycle, zeroes are written to RAM.?
A1I
74LS244
A1A
A1B
A1F
A1G
A1D
A1H
C7C
74LS21
A4B
B4B
B5B
A6B
B7B
SRAM1K4B
74LS244
74LS375
Konami GX319
Sheet: /Objects 2/
File: objects2.kicad_sch
Title: Mega Zone
Size: A3
Date: 2026-10-16
KiCad E.D.A. 10.0.4
Rev:
Id: 4/10

Timing



Active linebuffer toggles here

Pixel clock: 18.432MHz / 3 = 6.144MHz, T = 0.16276us
Hsync: Low 32px, High: 352px, Total: 384px, f = 6.144MHz / 384 = 16kHz
HBlank: Low 97px (Flipped 96). Front Porch: 32px, Back Porch: 32px,
Active pixels: 384 - 97 = 287 (288 flipped)

Diagram is for non-flipped timing.

Clocks

B16 18.432MHz
C16A 74LS107
C16B 74LS107
E10B 74LS240
E10H 74LS240
C17A 74LS240
G6A 74LS74
G4E 74LS240
G5D 74LS32

18.432MHz / 3 = 6.144MHz

PXCK1: f_CLK1 = f_M18 / 3

Horizontal Counters

C15 74LS161 counter
E10E 74LS240
E10C 74LS240
E10G 74LS240
B15C 74LS86
B15D 74LS86
B15A 74LS86
B14D 74LS04
B15B 74LS86
E9 74LS377 flip flops
C17E 74LS240
C17F 74LS240
E7A PROM32W8B
E8A PROM32W8B
B14F 74LS04
RA4 1k x 8

HF[128..2]

Sync

E10A 74LS240
B11A 74LS32
DE18D 74LS08
E10D 74LS240

Object Line Buffer

G4B 74LS240
F3D 74LS08
F3C 74LS08
G3C 74LS32
F2A 74LS164
G2A 74LS164
B17H 74LS244
B_A14F 74LS244
B17A 74LS244
B_A14G 74LS244

LBRST is HSYNC rising edge delayed for 76 pixels, active 4 pixels

LB1EN = Line buffer 1 Enable
Toggles, active every second frame.
Toggles 98 pixels after HSYNC goes active.

G5A 74LS32
G5B 74LS32
G4A 74LS240
C14B 74LS74

Object Data Timing

F4 74LS161
F5 74LS138 decoder
F3A 74LS08
G4G 74LS240
D7A 74LS20
G4D 74LS240
G4F 74LS240
CC13 470pF

Object Phase counters
Object Data Byte

Vertical Counters

B12 74LS161
B13 74LS161
C14A 74LS74
B14A 74LS04
C12A 74LS86
C12B 74LS86
C12C 74LS86
C12D 74LS86
C13A 74LS86
C13B 74LS86
C13C 74LS86
C13D 74LS86

Counter starts at 8+16=24
Counter starts at 32+64+128=224

Vertical Blanking

F7B 74LS10
G6B 74LS74
C17H 74LS240
B_B14H 74LS240
B_D14A 74LS240
C43 47n
C40 10n

B_D14 is D_15 in original PCB.
The position on PCB is D14.

Vertical blanking stop:
128 + 64 + 32 + 16 = 240
Start at:
240 + 32 - 256 = 16
Active lines: 240-16= 224V

Object Counters

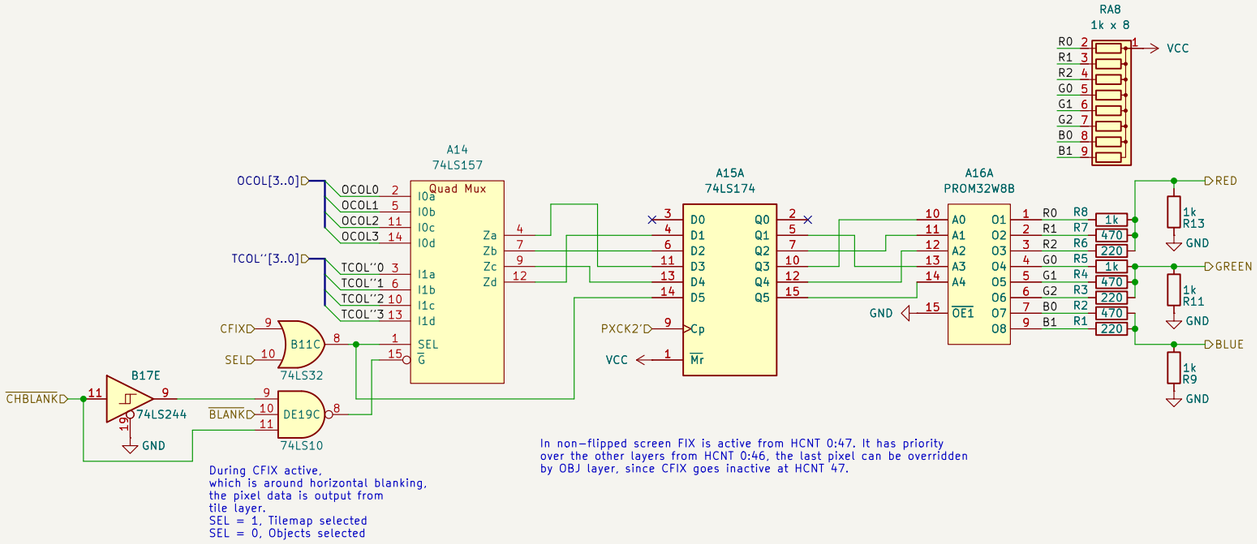
DMA is active for ~2.5 vertical lines.

G7A 74LS74
G7B 74LS74
F7C 74LS10
Quad Mux
A_D14 74LS161
A_D5 74LS157
A_D15 74LS161
G4C 74LS240
G4H 74LS240
F7A 74LS10
F6A 74LS74
F6B 74LS74

Counters start at zero although counter start value inputs are not connected.
When DMA is not active
- Counter goes from 0..143
When DMA is active
- Counter goes from 0..240

F7D 74LS10
F6C 74LS74
G7C 74LS74
E7B PROM32W8B
E8B PROM32W8B
C16C 74LS107
B15E 74LS86
C14C 74LS86
C12E 74LS74
C13E 74LS86
G6C 74LS74
E10I 74LS240
G4I 74LS240

Konami GX319	
Sheet: /Timing/	
File: timing.kicad_sch	
Title: Mega Zone	
Size: A3	Date: 2026-10-16
KiCad E.D.A. 10.0.4	**Rev:** Id: 5/10



Konami GX319

Sheet: /Color mixer/
File: color_mixer.kicad_sch

Title: Mega Zone

Size: A3	Date: 2026-10-16	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/10

$\overline{CE}$ = -($\overline{LATCHOE}$ & (~AB14 & ~AB15 & $\overline{MSHRAM}$))

Called CLK in original schematics

Called INT in original schematics

When $\overline{LACH}$ goes active, latch B_G9 goes inactive.

When $\overline{LACH}$ goes inactive, B_G9 latches value and drives it if CRAM or VRAM are still active.

0x0000..0x1FFF

0x2000..0x3FFF

Konami GX319		
Sheet: /Main CPU/		
File: mcpu.kicad_sch		
Title: Mega Zone		
Size: A3	Date: 2026-10-16	**Rev:**
KiCad E.D.A. 10.0.4		Id: 7/10

KONAMI-1: MA, MD
Z80: SA, DB
I8039: IA, ID

DB[7..0]
DAC[7..0]
ID[7..0]
IA[11..0]
YA[7..0]
SA[15..0]
SA[10..0]
CHA[1..0]
CHB[1..0]
CHC[1..0]

B_E7 Z80
B_B4 8039
B_A2A 74LS373
B_A3A 27C32
B_D14G 74LS240
B_D10A 74LS244
B_D10E 74LS244
B_D14D 74LS240
B_D14E 74LS240
B_C10A 74LS74
B_C11C 74LS08
B_C9A 74LS74
B_C9B 74LS74
B_C11D 74LS08
B_B12B 74LS32
B_DE9C 74LS32
B_RA6 4.7k x 4
B_D4 74LS374
B_B8A AY-3-8910
DE6A 27C64
B_B7 74LS138
B_A11D 74LS32
B_A11E 74LS04
B_DE12A 74LS32
B_DE11D 74LS02
B_A11A 74LS04
B_A11B 74LS04
B_A11C 74LS04
B_A11F 74LS04
B_A12A 74LS293
B_B11A 74LS293
B_B10A 74LS293
B_B9A 74LS293
B_A5A 4066
B_A5B 4066
B_A6A 4066
B_A6B 4066
B_A6C 4066
B_A6D 4066
B_A5C 4066
B_A5D 4066
B_B6A M5224P
B_B6B M5224P
B_B6C M5224P
B_B6D M5224P
B_B6E M5224P
B_A11G 74LS04
B_A5E 4066
B_A6E 4066
B_C11E 74LS08
B_C9C 74LS74
B_A2B 74LS373
B_C11A 74LS08
B_C11B 74LS08
B_B8B AY-3-8910
B_A3B 27C32
B_B9B AY-3-8910
B_B10B 74LS293
B_A12B 74LS293
B_B11B 74LS293
B_C3A LA4460
B_C3B LA4460

R47 47R
R49 470R
R50 470R
R51 820R
R52 820R
R53 1k
R40 1k
R41 1k
R42 1k
R43 10k
R44 10k
R45 10k
R46 10k
B_R24 200k
B_R25 200k
B_R26 200k
B_R27 200k
B_R28 200k
B_R29 200k
B_R30 200k
B_R31 200k
B_R32 100k
B_R33 100k
B_R34 100k
B_R35 100k
B_R36 100k
B_R37 100k
B_R38 100k
B_R39 200k
B_R48 4k7
B_VR200 180R
B_R1 4R7
B_R2 4R7
B_C1 10n
B_C2 10n
B_C3 10n
B_C4 100u
B_C5 100u
B_C6 10n
B_C7 10n
B_C8 10n
B_C9 1000u
B_C28 220n
B_C29 10n
B_C30 10n
B_C31 220n
B_C32 220n
B_C33 10n
C35 100p
X1 14.318MHZ

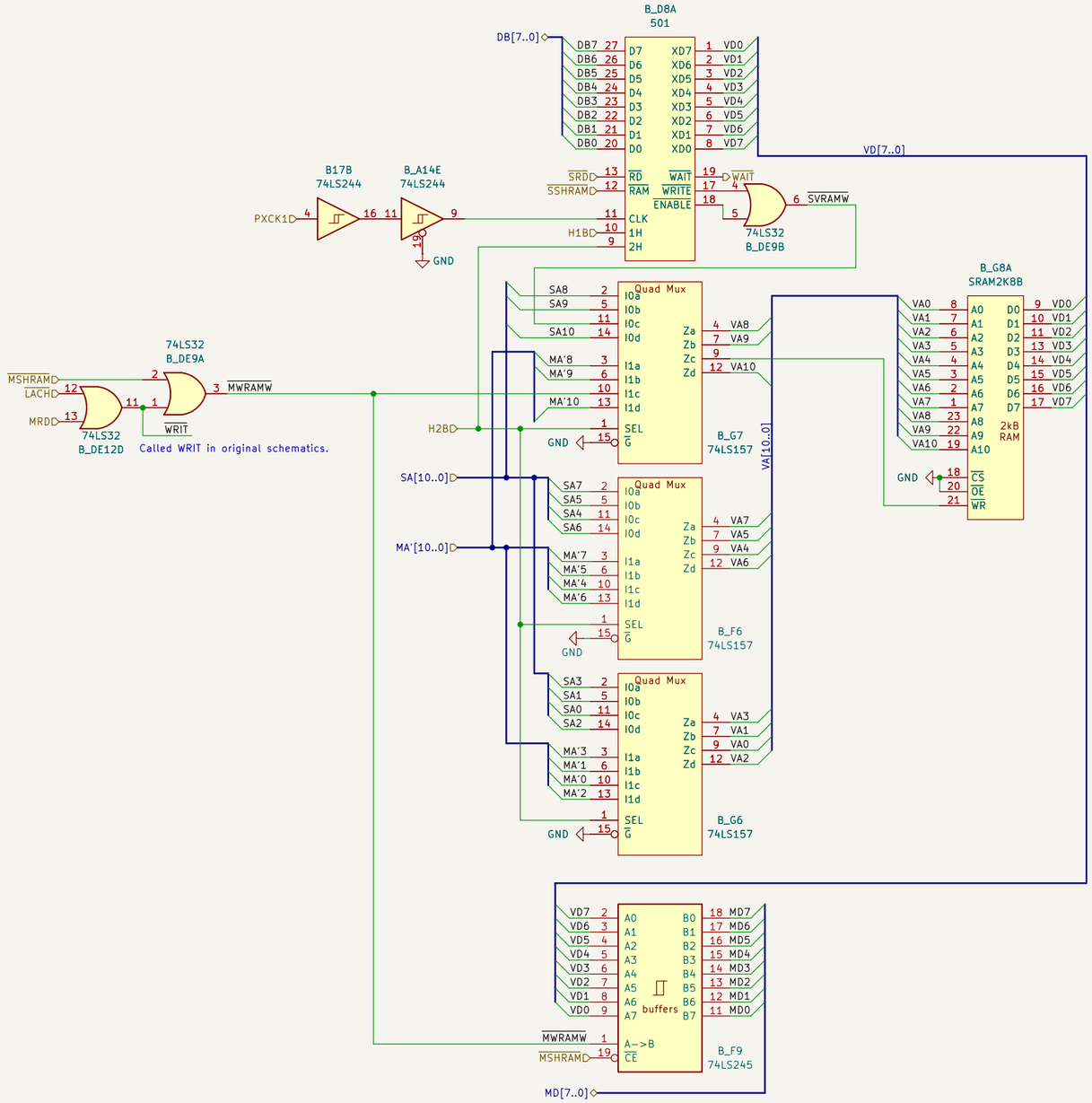
14.318MHz / 8 = 1.7897MHz
14.318MHz / 256 = 55.929687kHz
YA4 14.318MHz/8192 = 1747.8Hz
YA5 14.318MHz/16384 = 873.9Hz
YA6 14.318MHz/32768 = 437.0Hz
YA7 14.318MHz/65536 = 218.5Hz

Konami GX319	
Sheet: /Sub CPU/	
File: scpu.kicad_sch	
Title: Mega Zone	
Size: A3	Date: 2026-10-16
KiCad E.D.A. 10.0.4	Id: 8/10

Coin counters

Watchdog and reset

Konami GX319	
Sheet: /IO/	
File: io.kicad_sch	
Title: Mega Zone	
Size: A3	Date: 2026-10-16
KiCad E.D.A. 10.0.4	**Rev:** Id: 9/10



Konami GX319

Sheet: /Shared RAM/
File: shram.kicad_sch

Title: Mega Zone

Size: A3	Date: 2026-10-16	**Rev:**
KiCad E.D.A. 10.0.4		Id: 10/10